

Marking guide for the "Computing Ethics" final exam (BUC3216)

Question One

a. Define ethics, morals, and law and discuss the interrelationships between them. (12 Marks)

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- Ethics: Philosophical study of right and wrong conduct; a system of moral principles. (2 marks)
- Morals: Personal principles regarding right and wrong; shaped by culture, religion, and upbringing. (2 marks)
- Law: Rules created by government to regulate society and ensure justice; legally enforceable. (2 marks)

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Interrelationships:

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- Ethics may guide the creation of laws.
- Laws reflect societal moral standards.
- Something can be legal but unethical, and vice versa. (6 marks)

b. Two traditional ethical theories and how they guide decisions (13 Marks)

Utilitarianism (6.5 marks)

- Decision is judged by consequences – the greatest good for the greatest number.
- In product development: Ethical review ensures no harm; rushing may benefit profit but harm users.

Deontology (6.5 marks)

- Focus on duties, rules, and intentions rather than outcomes.
- The company has a duty to protect user data; ethical review is a moral obligation.

Question Two

a. Three moral problems encountered using the Internet (7 Marks)

Answer (any 3, 2 marks each + 1 for explanation):

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- Privacy violations
- Cyberbullying or harassment
- Distribution of explicit or harmful content
- Spam and misinformation

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b. Ethical dilemmas: censorship vs. freedom of speech; educational strategies (10 Marks)

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Dilemmas: Removing content may protect users but restrict expression (5 marks).

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Strategies:

- Group discussions foster awareness of diverse views.

- Scenario-based learning helps IT professionals anticipate challenges.
- Promotes critical thinking and ethical reflection. (5 marks)

c. Role of professional codes like ACM Code of Ethics (8 Marks)

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- Provide standards for ethical behavior.
- Encourage respect for privacy, avoidance of harm, and honesty.
- Guide developers in resolving conflicts like censorship vs. openness.
- Reinforce accountability and professionalism.

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Question Three

a. Define data privacy; types of personal info protected (8 Marks)

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Data privacy: Right to control access to one's personal information. (2 marks)

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Types protected:

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- Contact details
- Health info
- Location data
- Biometric info
- Financial records (6 marks)

b. Importance of data privacy today (10 Marks)

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- Builds trust with users and clients
- Reduces risks of identity theft, fraud, reputational damage
- Enhances regulatory compliance
- Essential for ethical responsibility in data-driven businesses

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c. Harmful outcomes of inadequate privacy (7 Marks)

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- Identity theft, financial fraud
- Loss of consumer trust
- Legal penalties
- Reputation damage
- The app's breach caused user backlash and legal scrutiny

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Question Four

a. Define intellectual property (IP) and its IT significance (10 Marks)

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IP Rights: Legal protections for creations of the mind.

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Significance:

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- Startups rely on IP for competitive edge.
- Prevents unauthorized use of software, branding.
- Encourages innovation by protecting original work.

b. File sharing, peer-to-peer and moral dilemmas (7 Marks)

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- These technologies ease copyright violations
- Dilemmas arise between information freedom vs. authors' rights
- Developers may unknowingly use pirated components

c. Fair use and its ethical implications (8 Marks)

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- Fair use: Limited legal use of copyrighted content for purposes like education, commentary
- For creators: risk of exploitation
- For users: must use content responsibly
- Balance needed between innovation and rights protection

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Question Five

a. Uganda's cyber law frameworks (10 Marks)

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- Computer Misuse Act: prohibits unauthorized access, cyber harassment
- Data Protection and Privacy Act: regulates collection, processing, and storage of personal data
- Both provide frameworks for secure IT practices

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b. Synergy and enforcement challenges (10 Marks)

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Work together: One ensures secure systems; the other safeguards personal rights

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Challenges:

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- Low awareness
- Limited technical capacity
- Cross-border crimes
- Enforcement gaps

c. Security vs. privacy balance and ethical concerns (5 Marks)

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- Surveillance must be proportionate
- Risks of government overreach
- Ethics demand transparency, consent, and oversight

Question Six

a. Ethical implications of surveillance programs (8 Marks)

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- Risks: Mass surveillance, violation of privacy
- Principles:
- Proportionality
- Accountability
- Transparency
- Legal oversight

b. Effects on public trust (7 Marks)

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- Privacy loss leads to fear, resentment
- Users may adopt countermeasures (e.g., encryption)
- Trust in government decreases, affecting cooperation

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c. Ten Commandments of Computer Ethics – impact on government (10 Marks)

- Do not snoop in others' files
- Do not use a computer to harm others
- Use computers with integrity
- Governments must respect ethical use
- Builds legitimacy and public confidence

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Question Seven

Short notes (Choose any $5 \times 5 = 25$ Marks). Each carries 5 marks:

a. Moral problems in professional codes

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- Conflicts between employer policies and ethical standards
- Vague guidelines may lead to misinterpretation

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b. Ethical responsibilities in cybersecurity

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- Protecting user data
- Reporting vulnerabilities
- Avoiding misuse of access

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c. White-hat vs. black-hat hacking

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- White-hats: ethical, authorized
- Black-hats: illegal, malicious intent
- Motivations and consequences differ

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d. Cybercrime and ethical implications

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- Fraud, phishing, identity theft
- Damages trust, economy
- Raises questions of justice and accountability

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e. Impacts of computerization on business

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- Job displacement
- Efficiency and speed
- Risk of dependency and security threats

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g. Codes for assisting the handicapped

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- Systems should support accessibility

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- Inclusive design principles
- Equal digital rights

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h. Seven principles of professional ethics

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- Public interest, integrity, competence, confidentiality, etc.
- Foundation for responsible conduct

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i. Patent vs. copyright vs. trademark

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- Patent: inventions
- Copyright: creative works ☐
- Trademark: brand identifiers
- Each protects different IP areas

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